Virtual Memory, Paging Architecture, and Page Tables

Virtual Memory decouples user logical memory from physical memory, allowing execution of processes that exceed available physical RAM. Paging divides logical address space into fixed-size Pages and physical RAM into Page Frames of identical size (typically 4 KB).

Address Translation: The CPU generates a logical address containing Page Number (p) and Offset (d). The Memory Management Unit (MMU) uses p as an index into the Page Table to retrieve Frame Number (f). Physical Address = (f * PageSize) + d. Page Table Entries contain valid/invalid bits indicating presence in RAM.

Translation Lookaside Buffer (TLB) and Page Faults

To avoid two memory accesses per address translation (one for the page table, one for physical data), hardware provides a Translation Lookaside Buffer (TLB)?a fast associative cache. Effective Memory Access Time (EMAT) = HitRate * (TLBAccess + MemAccess) + (1 - HitRate) * (TLBAccess + 2*MemAccess).

When a process accesses a page with valid bit = 0, a Page Fault trap is generated:
1. OS traps to kernel and checks if address was legitimate.
2. OS finds a free physical page frame (or evicts an existing page).
3. Disk controller initiates I/O to fetch page contents into frame.
4. Page table is updated with frame number and valid bit = 1.
5. Instruction causing the page fault is restarted.

Page Replacement Algorithms: FIFO, LRU, and Clock

When physical RAM is exhausted, page replacement algorithms choose a victim frame to swap out to disk:
1. FIFO: Replaces the oldest page. Suffers from Belady's Anomaly (allocating more frames can paradoxically increase page faults).
2. Optimal (OPT): Replaces the page that will not be used for the longest future time. Used as an unattainable theoretical benchmark.
3. Least Recently Used (LRU): Replaces the page that has not been referenced for the longest past interval. Free from Belady's Anomaly.
4. Clock Algorithm (Second Chance): Uses a circular buffer and a reference bit. If bit is 1, it resets bit to 0 and advances hand; if bit is 0, it selects that frame for replacement.